

Architecture Descriptor / Manifest

Applications, Topics, Brokers, Hosts,
Shared Libraries & QoS Contracts

Ingestion

Raw Structural Graph ($G_{\text{structural}}$)

Native typed topology

Typed projection
& QoS weights

Analysis Multigraph (G_{analysis})

Derived **DEPENDS_ON** edges & QoS-weighted
logical flows; typed node features

RANKING PATHWAY

Ranking Methods

- Analytical: dependency counts and first-order expansions (references); QoS-weighted betweenness (baseline)
- Hybrid: learned correction of the baseline
- Learned: GNN on DEPENDS_ON or multigraph; GBM on dependency features

Top-K Critical Component Set

- Predicted cascade impact $\hat{I}^*(v)$
- Ranking of unseen architectures

Evaluated against
simulated labels

Simulation Oracles (I^* , I_{dyn} , I_{comp})

Run on raw $G_{\text{structural}}$

Offline: training and evaluation labels

Independent run
on G_{struct}